INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 6.0

(DATA VISUALIZATION)

Project Title:

Infosys_HotelRevai AI Driven Revenue Analysis for Hotels

Name of the Intern **:**

CHELIKANI HARSHAL

Milestone 4 :

Visualization Storytelling

TABLE OF CONTENTS

1. Abstract
2. Introduction
3. Problem Statement
4. Objectives
5. Dataset Overview
6. Data Preprocessing
7. Data Modeling
8. DAX Metrics
9. Dashboard Overview
10. Business Impact
11. Conclusion
12. Future Scope
13. References

ABSTRACT

Revai is an AI-driven revenue analysis system designed specifically for the hospitality industry. This project leverages a comprehensive 4-year hotel booking dataset spanning 2014 to 2017 to uncover actionable business insights through systematic data preprocessing, advanced DAX metric modeling, and interactive Power BI dashboards. The system analyzes key performance indicators including total revenue, cancellation rates, booking behavior, and customer segmentation across City and Resort hotel types. By transforming raw transactional booking data into structured business intelligence, Revai empowers hotel management to make data-driven decisions that optimize revenue, reduce cancellation losses, and significantly improve operational efficiency.

INTRODUCTION

The hotel industry operates in a highly dynamic and competitive environment where revenue management and customer behavior analysis play a critical role in determining profitability. Traditional approaches to booking data analysis often fall short in providing real-time, granular insights that hotel managers need to act quickly and confidently.

Revai addresses this gap by building an end-to-end analytical pipeline — from raw data ingestion and cleaning, through KPI modeling, to fully interactive dashboard visualization. Using Microsoft Power BI and DAX (Data Analysis Expressions), this project translates complex booking-level transactional data into five purpose-built interactive dashboards covering revenue trends, booking behavior, customer segmentation, predictive cancellation risk, and detailed cancellation pattern analysis.

The project follows a structured workflow: raw booking data is first cleaned and modeled in Power Query Editor, enriched with calculated columns, and then powered by reusable DAX-based KPIs. The resulting dashboards enable hotel stakeholders at all levels to make strategic, informed decisions backed by data.

PROBLEM STATEMENT

Hotels generate vast volumes of booking data on a daily basis but frequently lack the analytical tools and frameworks to convert this data into meaningful revenue insights. This disconnect between data availability and actionable intelligence leads to significant operational and financial challenges.

The key challenges this project addresses are:

- **High cancellation rates** leading to revenue leakage, poor demand forecasting, and underutilized room inventory.
- **Lack of visibility** into which market segments and customer types are the most profitable and which are the highest risk.
- **Inability to identify pricing inefficiencies** due to fragmented and unstructured ADR (Average Daily Rate) analysis across time periods and hotel types.
- **No predictive framework** to proactively assess cancellation risk before it impacts occupancy and revenue targets.
- **Limited understanding** of customer behavior patterns across geographies, distribution channels, and booking lead times.
- **Absence of a unified dashboard** that consolidates all KPIs into one interactive, filterable view for management decision-making.

Revai solves these challenges by delivering a structured, KPI-driven analytics platform built on Power BI that provides hotel stakeholders with clear and actionable insights from historical booking data.

OBJECTIVES

The primary objectives of the Revai project are:

1. To clean, preprocess, and model raw hotel booking data into a structured, analysis-ready format.
2. To engineer meaningful calculated columns such as Total Stay, Total Guests, and Revenue to enrich the dataset.
3. To define reusable DAX-based KPIs covering revenue, cancellations, occupancy, and guest metrics.
4. To build five interactive Power BI dashboards that address distinct business questions across revenue, bookings, customer behavior, predictive risk, and cancellation analysis.
5. To enable data-driven decision-making for hotel management by surfacing

trends, anomalies, and segment-level insights.
6. To establish a scalable analytical framework that can be extended with real-time data or machine learning capabilities in future iterations.

DATASET OVERVIEW

The dataset used in this project is a publicly available 4-year Hotel Booking Dataset containing booking-level transactional records from 2014 to 2017. It includes data from two distinct hotel types: City Hotels and Resort Hotels.

Key attributes in the dataset include:

- Hotel type (City / Resort)
- Market segment and distribution channel
- Guest details — Adults, Children, Babies
- Stay details — Weekday nights and Weekend nights
- ADR (Average Daily Rate)
- Cancellation flag (is_canceled)
- Arrival date information — Year, Month, Day
- Country of origin
- Customer type
- Lead time (days between booking and arrival)
- Repeated guest flag
- Assigned and reserved room type

The dataset consists of granular booking-level records, making it ideal for KPI-driven analysis, segmentation studies, and trend identification across time periods, hotel types, and customer groups.

DATA PREPROCESSING

Before any analysis was performed, the raw dataset underwent thorough cleaning and transformation within Power Query Editor in Power BI.

Steps performed:

- **Removed unnecessary columns** that were irrelevant to the analysis scope, reducing noise and improving model performance.
- **Handled missing values** by identifying null or blank fields and applying

appropriate treatments such as filling with defaults or removing affected rows.

- **Removed duplicate records** to ensure the integrity and accuracy of booking counts and revenue calculations.
- **Corrected data types** to ensure consistency:
  - Date fields were converted to proper Date format.
  - Numeric fields (ADR, nights) were set to Whole Number or Decimal as appropriate.
  - Categorical fields (Market Segment, Customer Type) were set to Text.
- **Standardized categorical values** to ensure consistent grouping across fields such as Market Segment, Distribution Channel, and Customer Type, preventing fragmentation in visual outputs.

These preprocessing steps ensured that the downstream data model and DAX calculations operated on clean, reliable data.

DATA MODELING

To enrich the cleaned dataset and enable more meaningful KPI calculations, three new calculated columns were created within Power BI:

Total Stay Represents the total number of nights a guest stayed at the hotel.

Total Stay = Weekend Nights + Weekday Nights

Total Guests Represents the total number of guests in a booking, across all guest categories.

Total Guests = Adults + Children + Babies

Revenue Represents the estimated revenue generated from each booking.

Revenue = ADR × Total Stay

These calculated columns serve as the foundational building blocks for the DAX measures and dashboard KPIs defined in the next phase.

DAX METRICS

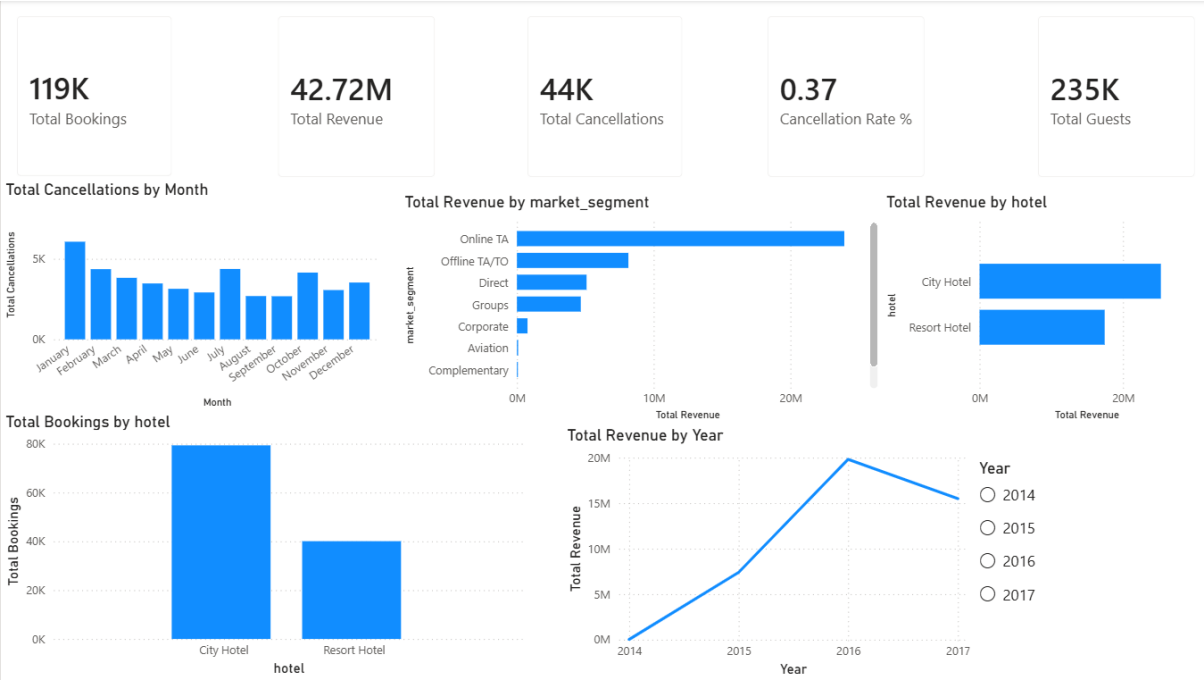
Following data modeling, DAX (Data Analysis Expressions) was used to define reusable, dynamic business KPIs. These measures power all five dashboards and can respond dynamically to filters and slicers applied by the user.

1. Average ADR
Average ADR = AVERAGE(hotel_booking_main[adr])
Used to understand pricing trends across time periods and hotel types.
2. Total Bookings
Total Bookings = COUNTROWS(hotel_booking_main)
Core business volume metric. Counts all booking records in the dataset.
3. Total Cancellations
Total Cancellations = CALCULATE(COUNTROWS(hotel_booking_main), hotel_booking_main[is_canceled] = 1)
Measures the total number of bookings that were cancelled.
4. Total Confirmed
Total Confirmed = CALCULATE(COUNTROWS(hotel_booking_main), hotel_booking_main[is_canceled] = 0)
Measures the total number of bookings that were successfully confirmed and not cancelled.
5. Cancellation Rate %
Cancellation Rate % = DIVIDE([Total Cancellations], [Total Bookings], 0)
Expresses cancellations as a proportion of total bookings. The third parameter (0) handles division-by-zero scenarios safely.
6. Total Guests
Total Guests = SUM(adults) + SUM(children) + SUM(babies)
Aggregates total guest volume across all bookings.
7. Total Revenue
Total Revenue = SUMX(hotel_booking_main, hotel_booking_main[adr] × (stays_in_week_nights + stays_in_weekend_nights))
Calculates cumulative revenue by multiplying ADR with total nights for each booking row using SUMX iteration.
8. Total Nights
Total Nights = SUM(stays_in_week_nights) + SUM(stays_in_weekend_nights)
Aggregates total nights stayed across all bookings.
9. Revenue TY (Time Intelligence)
Revenue TY = CALCULATE([Total Revenue], DateTable[Year] = YEAR(TODAY()))

Uses time intelligence to filter revenue to the current year, enabling year-on-year comparisons.

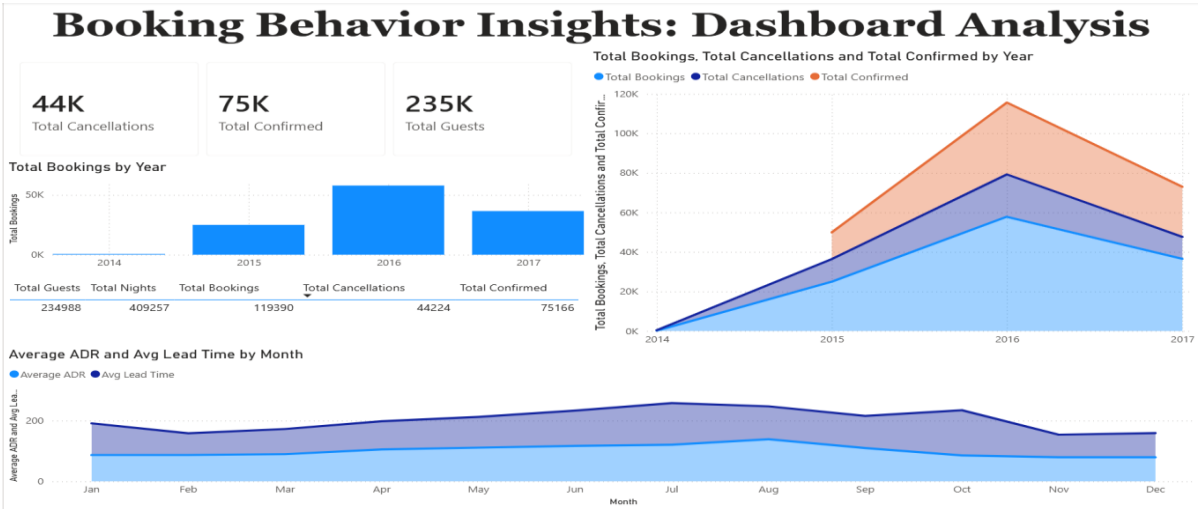
DASHBOARD OVERVIEW

Dashboard 1 — Revenue Dashboard



This dashboard provides a top-level view of hotel revenue performance. Key KPIs displayed include Total Bookings, Total Revenue, Total Cancellations, Cancellation Rate, and Total Guests. Visuals include revenue by market segment, year-wise revenue trends, monthly cancellation patterns, booking distribution across hotel types, and revenue comparison between City and Resort hotels. This dashboard helps management understand where revenue is coming from and which segments are most profitable.

Dashboard 2 — Booking Behaviour Analysis

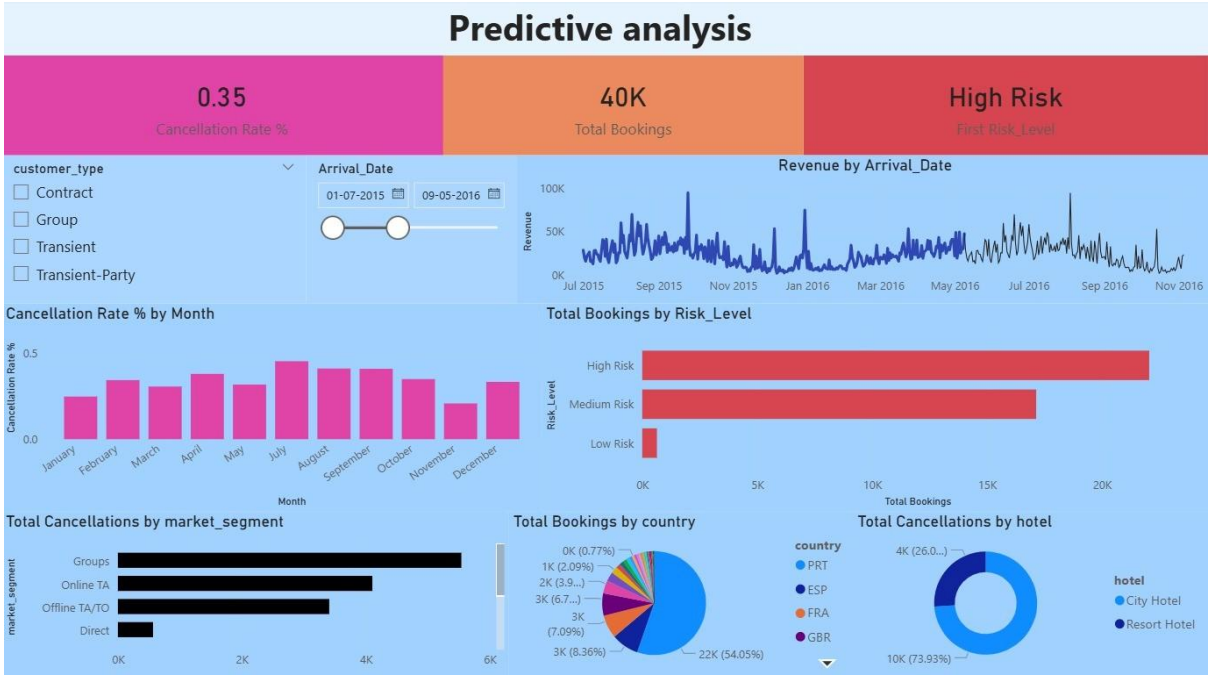


This dashboard focuses on booking patterns and lead time dynamics. Key visuals include total bookings, confirmations, and cancellations by year, along with Average ADR versus Average Lead Time plotted as a monthly trend. Business value includes identification of peak booking months, understanding how lead time influences pricing, and enabling demand forecasting and operational planning.

Dashboard 3 — Customer Behaviour Analysis

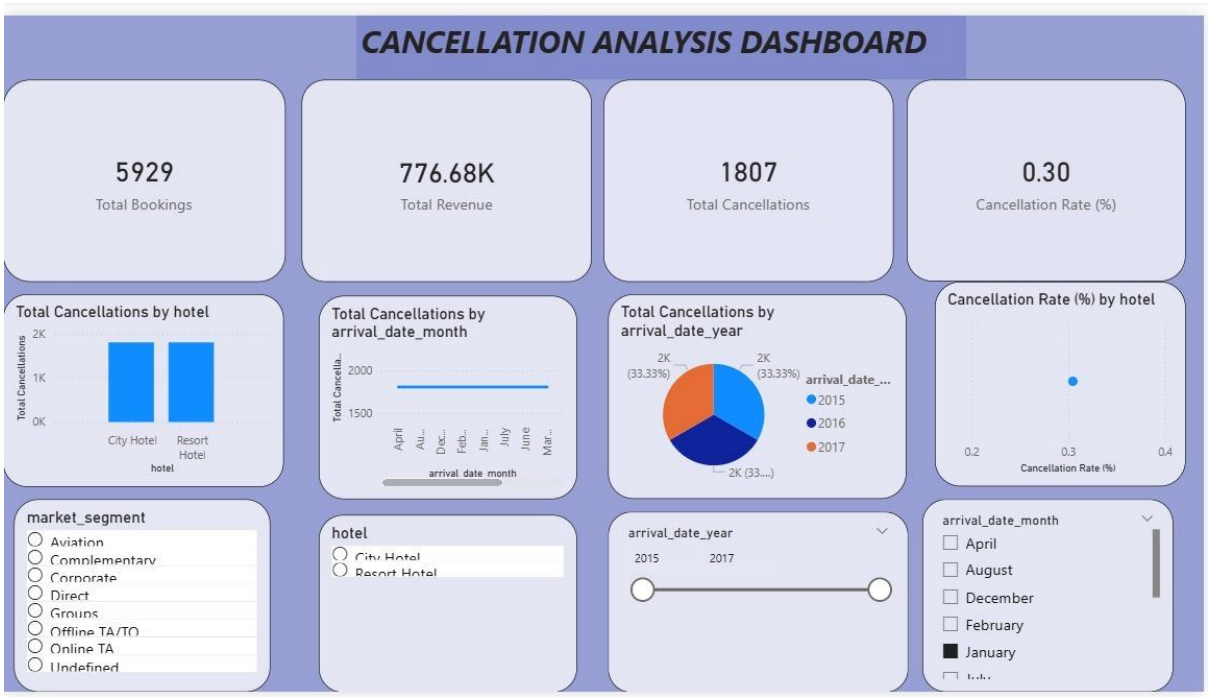
This dashboard enables deep customer segmentation and profiling. Key insights include bookings by market segment, cancellation rate by segment, guest composition (Adults, Children, Babies), customer type breakdown, and country-wise booking distribution. Business impact includes identifying high-risk cancellation segments, revealing the most profitable customer groups, and enabling targeted marketing strategies.

Dashboard 4 — Predictive Analysis Dashboard



This dashboard is focused on cancellation risk evaluation using historical patterns as a predictive proxy. Key KPIs include Cancellation Rate, Total Bookings, and Risk Level classification. Visuals include revenue by arrival date, cancellation rate by month, bookings segmented by risk level, cancellations by hotel type, bookings by country, and cancellations by market segment. This dashboard supports proactive risk management and resource allocation decisions.

Dashboard 5 — Cancellation Analysis Dashboard



This dashboard provides a granular breakdown of cancellation behavior. Key KPIs displayed include Total Bookings, Total Revenue, Total Cancellations, and Cancellation Rate (%). Visuals are designed to help management understand cancellation patterns by time period, hotel type, and customer segment, enabling targeted interventions to reduce revenue loss from cancellations.

BUSINESS IMPACT

The Revai project delivers measurable business value across five key dimensions:

1. Revenue Optimization Identified top-performing market segments and compared revenue performance between City and Resort hotels. ADR trends were analyzed across months and years to reveal pricing opportunities and seasonal patterns.

2. Improved Occupancy and Demand Planning Booking distribution was analyzed across years and months, lead time patterns were studied, and guest volume trends were evaluated — providing the foundation for accurate demand forecasting and occupancy optimization.

3. Risk Monitoring and Revenue Protection By measuring cancellation rates and identifying seasonal and hotel-level cancellation risks, the platform enables management to take proactive steps to protect revenue from cancellation leakage.

4. Customer Segmentation and Targeting Customers were segmented by market segment and customer type. Repeat guest percentages were identified, enabling more personalized marketing campaigns and loyalty strategies targeting high-value groups.

5. Data-Driven Decision Framework Reusable DAX-based KPIs and five interactive dashboards with dynamic filters provide a scalable decision-support framework that reduces reliance on intuition and enables consistent, evidence-based management decisions.

CONCLUSION

The Revai project successfully demonstrates how AI-driven data analytics can transform hotel revenue management. By processing and modeling a comprehensive 4-year booking dataset, the project delivered five interactive dashboards that provide deep visibility into revenue performance, booking trends,

customer behavior, and cancellation risk.

The DAX-powered KPIs and Power BI visualizations equip hotel management with the tools needed to optimize pricing strategies, reduce cancellation losses, and target high-value customer segments more effectively. The structured analytical pipeline — from raw data through preprocessing, modeling, and visualization — establishes a reusable and scalable framework.

This project not only addresses the immediate analytical needs of hotel revenue management but also lays the groundwork for future enhancements, including real-time data integration, machine learning-based cancellation prediction, and dynamic pricing recommendations.

FUTURE SCOPE

While Revai delivers significant value in its current form, several enhancements can further extend its capabilities:

- **Real-time data integration** — Connect the Power BI model to live booking systems via APIs or streaming datasets for up-to-the-minute dashboards.
- **Machine learning integration** — Build predictive models using Python or Azure ML to forecast cancellation probability at the individual booking level with greater accuracy.
- **Dynamic pricing recommendations** — Develop an AI-based pricing engine that suggests optimal ADR adjustments based on demand patterns, lead time, and segment behavior.
- **Sentiment analysis** — Incorporate guest review data to correlate satisfaction scores with booking and cancellation patterns.
- **Multi-property expansion** — Extend the framework to support analysis across multiple hotel properties with cross-property benchmarking.
- **Automated reporting** — Schedule automated report delivery to stakeholders via Power BI service subscriptions and email alerts.

REFERENCES

Dataset: Hotel Booking Demand Dataset (2014–2017) — publicly available booking-level transactional data.

Tool: Microsoft Power BI Desktop — for data modeling, DAX development, and dashboard visualization.

Tool: Power Query Editor — for data cleaning and transformation.
Language: DAX (Data Analysis Expressions) — for KPI and measure development.

Methodology: KPI-driven analytical dashboard design for hospitality revenue management.